

WMS

Workplace Management System

Complete Product Plan — Scratch to Rollout

STACK

.NET 8 + Angular 17+

ARCHITECTURE

Modular Monolith

DATABASE

PostgreSQL

TARGET

Indian SME / Enterprise

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SECTION 1

Executive Summary

WMS (Workplace Management System) is a multi-tenant SaaS platform built on .NET 8 and Angular 17+ targeting Indian SMEs and mid-market enterprises with 50–500 employees. It provides a unified system for employee lifecycle management, asset tracking, attendance, bookings, compliance, and HR operations — the feature set that tools like Keka, DarwinBox, and GreytHR are built on, delivered at a competitive price point with a superior onboarding experience.

Target Market	Indian SME / Mid-market (50–500 employees)
Primary Buyers	HR Managers, Operations Heads, Company Founders
Competitive Edge	3-minute AI-powered onboarding, clean UX, affordable pricing
Revenue Model	Monthly SaaS subscription, tiered by employee count
Launch Target	Phase 1+2 MVP in 14–16 weeks
Stack	.NET 8 Web API + Angular 17+ + PostgreSQL + Docker + Azure

The single biggest differentiator: AI-powered company setup. A new company admin can register, pick their industry, and have departments, designations, asset types, and leave policies pre-populated in under 3 minutes. No competitor offers this.

SECTION 2

System Overview & Vision

WMS is designed as a Modular Monolith — a single deployable unit with internally decoupled modules that can be split into microservices later without redesign. Every module is independently testable, owns its domain, and communicates via MediatR events internally.

IDENTITY • Company registration • Multi-tenancy • JWT auth • Role management	EMPLOYEES • Profiles & lifecycle • ID card + QR • Invite flow • Org chart	ASSETS • Asset register • Assignment • Service requests • Lifecycle
ATTENDANCE • Clock in/out • Leave management • Timesheets • Shifts	BOOKINGS • Meeting rooms • Desk booking • Visitor mgmt • Parking	ANALYTICS • Dashboards • Reports • Exports • Audit log

Multi-tenancy principle

Every table in the database carries a CompanyId foreign key. Every API endpoint is automatically scoped to the authenticated user's company. No query ever returns data across company boundaries. This is enforced at the repository layer, not the controller.

Architecture rule: All domain entities inherit from BaseEntity which carries Id, CompanyId, CreatedAt, UpdatedAt, CreatedBy, IsDeleted. Soft deletes only — nothing is ever hard deleted.

SECTION 3

Roles & Permissions

Role	Who they are	Key permissions
Super Admin	Platform owner (you)	Manage all companies, billing, platform config
Company Admin	Company founder/owner	Register company, manage HR users, billing
HR Manager	HR department staff	Add/edit employees, manage assets, approve leaves
Manager	Team leads	View team, approve leave & timesheet, assign tasks
Employee	All staff	View own profile, assets, request service/assets, clock in/out

Roles are stored in the database and loaded on JWT issuance. Permission checks use policy-based authorization in .NET 8. Future versions will support custom role creation per company.

Key rule: Employees can only READ their own data. Only HR can WRITE employee records. Managers can READ their team's data. Company Admin cannot edit employee records — that is strictly HR territory.

PHASE 1

Company Setup & Employee Management

6–8 weeks

Duration
6–8 weeks

PHASE 1 · SECTION 4.1

Company Registration & AI Onboarding

The onboarding wizard is the most important UX in the entire product. A company admin completes 5 steps and has a fully working system in under 3 minutes.

Step	Screen	What happens
1	Company basics	Name, domain, logo upload, country, timezone, GST number
2	Industry & size	Pick industry from list — AI instantly suggests full structure
3	Review & customize	Edit departments, designations, asset types — all editable cards
4	First HR user	Name + email → invite email auto-sent with set-password link
5	Done!	Dashboard loads — 'Import employees via Excel' CTA is prominent

AI suggestion engine

When the admin picks an industry, the system pre-fills:

Industry	Departments suggested	Asset types suggested
IT / Software	Engineering, QA, DevOps, Product, Design, HR, Finance, Sales	Laptop, Monitor, Phone, Headset, Access Card
Manufacturing	Production, QC, Maintenance, Logistics, HR, Finance	Safety Helmet, Tools, Machinery, Vehicle
Healthcare	Clinical, Nursing, Admin, Pharmacy, HR, Finance	Medical Equipment, Laptop, Access Card
Retail	Store Ops, Inventory, Sales, Customer Service, HR	POS Terminal, Laptop, Vehicle
Education	Academic, Admin, IT, Finance, HR, Library	Laptop, Projector, Lab Equipment

Implementation: Phase 1 uses static JSON templates per industry (fast to ship, zero latency). Phase 7 upgrades this to a live Claude/GPT API call that generates custom suggestions based on company name, size, and country context.

PHASE 1 · SECTION 4.2

HR & Employee Management

HR CAN DO

- Add new employee
- Edit any employee field
- Change department / role
- Suspend / reactivate account
- Initiate offboarding
- Download ID card
- Bulk import via Excel

EMPLOYEE CAN DO

- View own profile (read-only)
- View own ID card
- Update emergency contact
- View org chart
- View company announcements
- Change own password

Employee lifecycle states

State	Description	Trigger
Active	Normal working employee	Default on creation
On Notice	Serving notice period	HR sets resignation date
Suspended	Temporarily deactivated	HR manual action
Offboarded	Exited the company	Exit clearance completed

Records are never deleted — only soft-deleted with `IsDeleted = true`. Full history is preserved for audit and compliance.

PHASE 1 · SECTION 4.3

Employee ID Card with QR Code

When HR saves a new employee, the system instantly generates and displays their ID card in the UI. The card is rendered as an Angular component and can be exported as PNG or PDF.

Field	Source	Notes
Employee ID	Auto-generated	Format: EMP-{YYYY}-{XXXX} e.g. EMP-2025-0042
Full name	Employee record	Displayed prominently
Designation	Employee record	Shown below name
Department	Employee record	Badge style
Join date	Employee record	Formatted as DD MMM YYYY
Company name + logo	Company settings	White-labeled per company
QR code	Generated from Employee ID	Links to: <code>/employees/{id}</code> (auth-gated)
Photo	Optional upload	Initials avatar if no photo

Security note: The QR code encodes a URL only — not raw employee data. Scanning requires authentication. Use qrcode.js on the Angular side and generate the unique URL from the employee GUID.

PHASE 1 · SECTION 4.4

Authentication & Invite Flow

Step	Actor	Technical detail
1. HR adds employee	HR	Employee record created with Status = Pending
2. Token generated	System	Guid token stored in InviteTokens table with 48hr expiry
3. Email sent	System (SendGrid)	Invite link: /auth/set-password?token={guid}
4. Employee clicks link	Employee	Frontend validates token against API

5. Se t p a s s w o r d	Employee	BCrypt hashed, token marked as Used + expired
6. Lo gin	Employee	JWT + Refresh token issued, role claims loaded
7. To ke n r e f r e s h	System	Refresh token rotated every 15 min (sliding window)

PHASE 2

Asset Management

5–6 weeks

Duration
5–6 weeks

PHASE 2 · SECTION 5.1

Asset Types & Register

HR first defines asset types (categories), then registers individual assets against those types. Asset types are seeded from the AI onboarding template and can be customized.

Entity	Fields
AssetType	Name, Description, Icon, CompanyId
Asset	AssetTypeId, Name, SerialNumber, Model, Brand, PurchaseDate, PurchaseCost, DepreciationRate, Condition, Status, CompanyId

PHASE 2 · SECTION 5.2

Assignment & Lifecycle

Status	Meaning	Transition trigger
Available	In stock, not assigned	Default / after return
In Use	Assigned to employee	HR assigns asset
Awaiting Acknowledgement	Assigned, not yet confirmed	Between assign and employee confirm
Under Service	Being repaired/serviced	Service request approved
Retired	Decommissioned	HR marks as retired

Critical enterprise requirement: Employee must digitally acknowledge receipt of an asset before status moves from 'Awaiting Acknowledgement' to 'In Use'. This creates a legal audit trail for asset handover.

PHASE 2 · SECTION 5.3

Employee Asset Requests

Request type	Fields	Approval flow
Service request	Asset, issue description, priority, photo (optional)	HR reviews → approves/rejects → asset status → Under Service
New asset request	Asset type needed, business justification, urgency	HR reviews → approves (assigns) / rejects with reason

PHASE 3

Attendance & Time Management

5–7 weeks

Duration
5–7 weeks

SECTION 6

Attendance & Leave Management

ATTENDANCE FEATURES

- Clock in / clock out
- Overtime auto-detection
- Late arrival flagging
- Work-from-home marking
- Monthly attendance summary
- Manager attendance view

LEAVE MANAGEMENT

- Leave request + approval
- Leave type config (CL/SL/EL/LWP)
- Leave balance tracking
- Holiday calendar (national + custom)
- Comp-off tracking
- Leave encashment support

Timesheet module

Employees submit weekly timesheets with project/task breakdown. Managers approve or reject with comments. Finance can run cost-center reports from timesheet data.

Shift management

Define shift templates (Morning 9–6, Night 10pm–6am, Rotational). Assign shifts to employees individually or by department. System flags attendance anomalies based on expected shift.

PHASE 4

Bookings & Facilities

4–5 weeks

Duration
4–5 weeks

SECTION 7

Bookings & Facility Management

Module	Features	Key rules
Meeting room booking	Book rooms by time slot, add attendees, calendar view	Conflict detection — double booking prevented
Desk / hot-desking	Floor plan view, book a desk for day/week	Capacity limits, preference saving
Parking slot booking	Assign/book parking bays	First-come basis or allocated
Visitor management	Pre-register visitors, OTP entry, badge print	Host notification on arrival
Cafeteria booking	Pre-book meal slots	Reduces cafeteria congestion

PHASE 5

Analytics, Reporting & Notifications

4–5 weeks

Duration
4–5 weeks

SECTION 8

Analytics & Reporting

The intelligence layer — turns raw data into actionable insight for HR and management.

Report	Audience	Format
Headcount by dept over time	HR / Finance	Chart + Excel
Joiners & leavers monthly	HR / Management	Table + PDF
Attendance heatmap	HR / Managers	Visual chart
Leave pattern analysis	HR	Chart + Excel
Asset register (auditor-ready)	Finance / Audit	PDF with logo
Asset utilization & depreciation	Finance	Excel
Cost center breakdown	Finance	Excel
Department-wise asset value	Finance	Chart + Excel

Notification Engine

Trigger	Recipients	Channel
Employee invite sent	New employee	Email
Asset assigned	Employee	Email + In-app
Service request status change	Employee	Email + In-app
Leave request submitted	Manager / HR	Email + In-app
Leave approved/rejected	Employee	Email + In-app
Contract end approaching (30d)	HR + Manager	Email
Asset acknowledgement pending	Employee	Email + In-app
New asset request submitted	HR	Email + In-app

PHASE 6

HR, Compliance & Document Management

6–8 weeks

Duration
6–8 weeks

SECTION 9

HR Operations

ONBOARDING WORKFLOW

- Checklist per employee
- IT setup tracking
- Document submission status
- Policy acknowledgement
- Training completion
- Access card issuance

EXIT CLEARANCE WORKFLOW

- Asset return checklist
- Knowledge transfer tasks
- Access revocation
- Final settlement trigger
- Experience letter generation
- Exit interview form

Document Vault

Each employee has a secure document vault. HR uploads documents; employees can view their own. All uploads are stored in Azure Blob Storage with SAS token access.

Document type	Uploaded by	Visible to
Offer letter	HR	Employee + HR
NDA / contracts	HR	Employee + HR
Aadhaar / PAN copy	Employee (self-serve)	HR only
Appraisal letters	HR	Employee + HR
Warning letters	HR	HR only
Experience letter	System (generated)	Employee + HR

Compliance

- Full audit log on every entity — field-level change tracking with user + timestamp
- DPDP Act (India) compliance — employee right to export their data, right to correction
- Soft deletes only — no hard deletion, full history preserved
- Policy acknowledgement with digital signature and timestamp
- Role-based data visibility — employees cannot see colleagues' data

PHASE 7

AI Features & Integrations

Ongoing

Duration
Ongoing

SECTION 10

AI-Powered Features

Feature	How it works	Value
Smart onboarding (Phase 1 upgrade)	Live Claude/GPT API call with company context	Custom dept/role suggestions per company
HR chatbot	RAG over company policies + employee data	Employees self-serve HR queries 24/7
Leave prediction	ML model on attendance history	HR can plan coverage proactively
Attendance anomaly detection	Flag unusual patterns	Reduces buddy-punching and fraud
Resume parser (onboarding)	Extract fields from CV PDF automatically	Faster employee data entry
Smart asset recommendation	Suggest asset type based on role	Faster asset assignment for HR

Key Integrations

Integration	Purpose	Priority
SendGrid / SMTP	Transactional emails (invites, notifications)	Phase 1 — critical
Azure Blob Storage	Document vault, ID card export storage	Phase 1
Google OAuth / Azure AD SSO	Single sign-on for enterprise clients	Phase 3
Google Calendar / Outlook	Sync meeting room bookings	Phase 4
Slack / MS Teams webhook	Real-time notifications in team channels	Phase 5
Razorpay / Stripe	SaaS subscription billing	Pre-launch
WhatsApp Business API	Critical HR alerts via WhatsApp	Phase 6
Anthropic / OpenAI API	AI features (chatbot, suggestions)	Phase 7

SECTION 11

Database Schema — Multi-tenant Design

All tables extend BaseEntity. CompanyId is on every table and all queries are scoped to it at the repository layer using a global query filter in EF Core.

Core / Identity module

Table	Key fields
Companies	Id, Name, Domain, LogoUrl, Industry, PlanId, IsActive, CreatedAt
Users	Id, CompanyId, Email, PasswordHash, RoleId, IsActive, LastLogin
Roles	Id, CompanyId, Name, IsSystemRole
Permissions	Id, RoleId, Module, CanRead, CanWrite, CanDelete
InviteTokens	Id, CompanyId, UserId, Token(Guid), ExpiresAt, UsedAt
RefreshTokens	Id, UserId, Token, ExpiresAt, IsRevoked
AuditLogs	Id, CompanyId, UserId, Entity, EntityId, Action, OldValue, NewValue, Timestamp

Employee module

Table	Key fields
Employees	Id, CompanyId, UserId, EmployeeCode, FirstName, LastName, DeptId, DesignationId, ManagerId, JoinDate, Status, PhotoUrl
Departments	Id, CompanyId, Name, HeadEmployeeId, ParentDeptId
Designations	Id, CompanyId, DeptId, Title, Level
EmergencyContacts	Id, EmployeeId, Name, Relationship, Phone
EmployeeDocuments	Id, EmployeeId, Type, FileUrl, UploadedBy, VerifiedAt
OnboardingChecklists	Id, EmployeeId, TaskName, IsCompleted, CompletedAt, CompletedBy
ExitRequests	Id, EmployeeId, ResignDate, LastWorkingDay, Reason, Status, ClearanceStatus

Asset module

Table	Key fields
AssetTypes	Id, CompanyId, Name, Description, IconUrl

Assets	Id, CompanyId, AssetTypeId, Name, SerialNo, Model, Brand, PurchaseDate, PurchaseCost, DepreciationRate, Status, Condition
AssetAssignments	Id, CompanyId, AssetId, EmployeeId, AssignedAt, AssignedBy, AcknowledgedAt, ReturnedAt
AssetServiceRequests	Id, CompanyId, AssetId, EmployeeId, IssueDescription, Priority, Status, ResolvedAt
AssetRequests	Id, CompanyId, EmployeeId, AssetTypeId, Justification, Urgency, Status, ReviewedBy, ReviewNote

Attendance module

Table	Key fields
AttendanceRecords	Id, CompanyId, EmployeeId, Date, ClockIn, ClockOut, WorkingHours, Status(Present/Absent/WFH/Leave)
LeaveTypes	Id, CompanyId, Name, TotalDays, IsCarryForward, IsPaid
LeaveBalances	Id, CompanyId, EmployeeId, LeaveTypeId, Year, TotalDays, UsedDays, PendingDays
LeaveRequests	Id, CompanyId, EmployeeId, LeaveTypeId, FromDate, ToDate, Reason, Status, ApprovedBy
Holidays	Id, CompanyId, Name, Date, IsOptional
Shifts	Id, CompanyId, Name, StartTime, EndTime, IsNightShift
Timesheets	Id, CompanyId, EmployeeId, WeekStartDate, Status, SubmittedAt, ApprovedBy

SECTION 12

Tech Stack & Architecture

Layer	Technology	Version	Why
Backend API	.NET Web API	8 (LTS)	Fast, modern, enterprise-grade
Architecture	Modular Monolith + Clean Arch	—	Scale without microservice complexity
CQRS	MediatR	12+	Clean separation of commands/queries
ORM	Entity Framework Core	8	Type-safe, migrations, global filters
Database	PostgreSQL	16	Open source, robust, EF support
Auth	JWT + Refresh tokens	—	Stateless, scalable
Email	SendGrid	—	Reliable transactional email
File storage	Azure Blob Storage	—	Secure, scalable, SAS tokens
API Docs	Scalar / Swagger	—	Modern API documentation
Frontend	Angular	17+	Enterprise-grade, opinionated
State management	NgRx Signals	—	Modern reactive state
UI components	Angular Material / PrimeNG	—	Consistent component library
QR code	qrcode.js (Angular)	—	Client-side QR generation
PDF export	jsPDF + html2canvas	—	ID card and report export
Containerization	Docker + Docker Compose	—	Consistent environments
Cloud	Azure App Service + Azure DB	—	.NET native, easy setup
CI/CD	GitHub Actions	—	Free, integrates with Azure
Monitoring	Azure App Insights	—	Performance + error tracking

Folder structure — backend

WMS.API – Controllers, middleware, program.cs
WMS.Application – CQRS handlers, DTOs, interfaces, validators
WMS.Domain – Entities, enums, domain events
WMS.Infrastructure – EF Core, repos, email, blob, external services
WMS.Modules/
Identity/ – Company, users, roles, auth
Employees/ – Profiles, lifecycle, ID card
Assets/ – Types, register, assignments, requests
Attendance/ – Clock, leave, timesheets
Bookings/ – Rooms, desks, visitors
Analytics/ – Reports, dashboards
WMS.SharedKernel – BaseEntity, common interfaces, exceptions

SECTION 13

Security Model

- **Multi-tenant isolation**

CompanyId global query filter in EF Core — impossible to query across companies accidentally

- **Password hashing**

BCrypt with work factor 12 — never store plain text

- **JWT security**

Short-lived access tokens (15 min) + long-lived refresh tokens (7 days, rotated on use)

- **Invite tokens**

One-time GUID tokens with 48hr expiry, single use, invalidated on use

- **Role-based access**

Policy-based authorization in .NET — every endpoint has explicit role requirement

- **QR code safety**

QR encodes URL only, not employee data. URL requires authentication to access

- **File access**

Azure Blob SAS tokens — time-limited, per-file signed URLs

- **API rate limiting**

ASP.NET Core rate limiting middleware — prevent brute force

- **HTTPS only**

HSTS enforced, HTTP redirect to HTTPS always

- **Audit trail**

Every create/update/delete logged with userId, timestamp, old+new values

- **Input validation**

FluentValidation on all commands — no raw SQL, parameterized queries only

- **CORS policy**

Whitelist only known frontend domains

SECTION 14

SaaS Monetization Strategy

Target buyer: HR Manager or Operations Head at a 50–500 employee Indian company. They are on spreadsheets or struggling with expensive legacy tools. Your sweet spot is undercutting Keka (₹6,000+/mo) with a better onboarding experience.

Plan	Price	Employees	Key features
Free	₹0 / mo	Up to 10	Phase 1+2 features, no exports, no custom branding
Starter	₹2,999 / mo	Up to 50	All Phase 1–3, email notifications, Excel import, ID card export
Growth	₹7,999 / mo	Up to 200	All phases, analytics, document vault, Google SSO, API access
Enterprise	Custom	200+	Custom branding, dedicated support, SLA, on-prem option, HRMS integration

Revenue projections (conservative)

Milestone	Customers	Mix	MRR
3 months post-launch	5 paying	3 Starter + 2 Growth	₹24,997
6 months	20 paying	12 Starter + 8 Growth	₹99,988
12 months	60 paying	35 Starter + 20 Growth + 5 Enterprise	₹4.2L+
24 months	200 paying	Scale with referrals + content marketing	₹15L+

Growth levers

- Free tier as a trojan horse — companies onboard free, grow past 10 employees, forced to upgrade
- Referral program — HR managers talk to other HR managers at networking events
- Content SEO — 'how to manage employee assets in India', 'HR compliance India 2025'
- Product Hunt launch — gets you initial paying users and backlinks
- Direct outreach — LinkedIn to HR Heads at 50–200 employee companies in Gurugram/Bangalore
- CA / HR consultant partnerships — they recommend tools to their clients

SECTION 15

Rollout & Go-to-Market

Stage	When	Goal	Success metric
Alpha	Weeks 1–14	Build Phase 1+2, internal testing	Zero critical bugs in happy path
Private Beta	Weeks 15–18	5 friendly companies onboard free	Onboarding < 5 min, daily active use
Public Beta	Month 5	Open free tier, collect feedback	50+ companies, NPS > 40
Paid Launch	Month 6	Enable billing, first paid conversions	5 paying customers
Growth	Month 7–12	Content, referrals, direct sales	20+ paying, MRR █1L
Scale	Year 2	Phase 5–7 features, enterprise deals	200 companies, █15L MRR

Pre-launch checklist

- Deployed to Azure with custom domain + SSL
- Razorpay subscription billing integrated and tested
- Email deliverability tested (SPF, DKIM, DMARC configured)
- Privacy policy + Terms of Service pages live
- DPDP Act compliance review done
- Load tested to 100 concurrent users
- Error monitoring (App Insights) configured
- Backup strategy configured (daily automated DB backups)
- Demo company account with sample data ready for sales demos
- Onboarding takes under 3 minutes — tested with 5 real people

SECTION 16

Timeline Summary

Phase	Module	Duration	Cumulative
Phase 1	Company setup, AI onboarding, employee mgmt, ID card, auth	6–8 weeks	8 weeks
Phase 2	Asset management, assignments, requests	5–6 weeks	14 weeks
Phase 3	Attendance, leave, timesheets, shifts	5–7 weeks	21 weeks
Phase 4	Meeting rooms, desk booking, visitor mgmt	4–5 weeks	26 weeks
Phase 5	Analytics, reporting, notifications, audit	4–5 weeks	31 weeks
Phase 6	HR ops, document vault, compliance, offboarding	6–8 weeks	39 weeks
Phase 7	AI features, SSO, integrations, mobile PWA	Ongoing	~1 year
—	Private beta, billing, public launch	4 weeks (parallel)	~6 months to paid launch

Recommended approach: Ship Phase 1 + Phase 2 as your MVP. Deploy it. Get 5 companies using it free. The feedback you get in weeks 15–18 will be worth more than 6 months of solo planning. Build Phase 3 onward based on what real users ask for most.

Final advice

- One well-built phase beats three half-built ones — resist scope creep until Phase 1+2 is solid
- Your modular monolith foundation means every module you add gets easier, not harder
- The AI onboarding wizard is your demo weapon — make it flawless before showing anyone
- Multi-tenancy and security are not afterthoughts — they are baked into every table and query
- You are 23 — this project alone, if live and deployed, will 3x your salary options in 18 months

End of document